

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)	[The <i>activity</i> is the] rate of decay or [the number of] disintegrations / decays per second (or unit time [of a sample of radioactive nuclei] (1) Accept number of radioactive emissions per second Becquerel or becquerel or Bq (1) Accept s ⁻¹	2			2		
		(ii)	$\lambda = \frac{\ln 2}{3.3 \times 10^5} = 2.1 \times 10^{-6} \text{ s}^{-1}$ (1) $e^{\lambda t} = \frac{A_0}{A}; \quad t = \frac{1}{\lambda} \ln \left(\frac{A_0}{A} \right)$ (1) $t = \frac{1}{(2.1 \times 10^{-6})} \ln \left(\frac{1}{0.2} \right) = 7.7 \times 10^5 \text{ s} = [8.9 \text{ day}]$ (1) Alternative: Taking logs of $A = \frac{A_0}{2^n}$ e.g. $\ln A = \ln A_0 - n \ln 2$ or $n \ln 2 = \ln \frac{A_0}{A}$ (1) Substitution: $n \ln 2 = \ln \frac{100}{20}$ [leads to $n = 2.32$] (1) Correct answer $7.7 \times 10^5 \text{ s} = [8.9 \text{ day}]$ (1)		3		3	3	
		(iii)	Safe outside the body or stopped by the skin or danger or dangerous if swallowed / inhaled (1) <u>Highly</u> ionising (1) Any reference to damage e.g. damage DNA, kills cells, cause cancer, cause radiation poisoning, causes mutations (1)			3	3		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(b)		Attempt at masses: LHS – RHS i.e. 0.005 229 [u] or 0.004131 [u] or 0.006327 [u] (1) Attempt at conversion to MeV or mass i.e. $\times 931$ or $\times 1.66 \times 10^{-27}$ (1) Attempt at conversion to J i.e. $\times 1.6 \times 10^{-13}$ or $\times c^2$ (1) Expect 7.8×10^{-13} [J] Rearrangement and incorporating 98 % or statement ignoring 98 % i.e. $v = \sqrt{\frac{2E}{m}}$ (1) N.B. Accept 4 u or 4.0026 u or 4.0015 u 1.52×10^7 or 1.35×10^7 or 1.67×10^7 [ms ⁻¹] ecf on 98 % and one arithmetic slip (1)	1 1	1 1 1		5	5	
			Question 1 total	4	6	3	13	8	0